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GSNSR: Improving Neural Surface Reconstruction with Geometric and Color Priors from 3DGS

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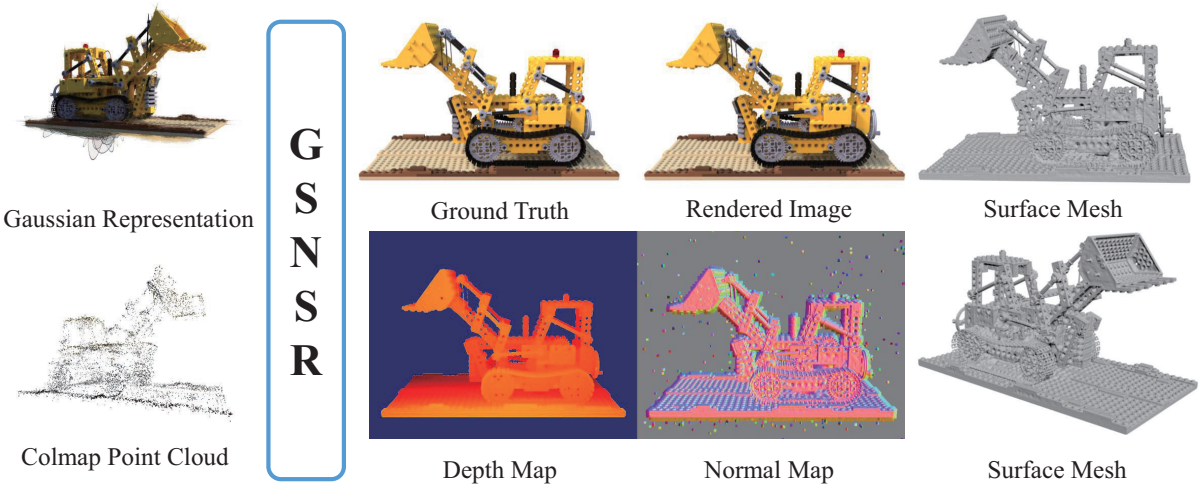


Figure 1: *Surface reconstruction by GSNSR on Lego and Mic from the NeRF Synthetic dataset. Taking a pretrained Gaussian model and a Colmap point cloud as inputs (left), GSNSR progressively refines geometry across training stages, with the final reconstruction shown as rendered RGB, depth, and normal maps (center) alongside the extracted mesh (right). For Lego, the reconstructed mesh accurately recovers holes without introducing artifacts, while for Mic, GSNSR successfully reconstructs the high-frequency mesh structure with fine geometric fidelity.*

Abstract

Neural implicit surfaces and 3D Gaussian Splatting (3DGS) bring complementary strengths to 3D reconstruction, with implicit surfaces yielding coherent geometry and consistent topology while 3DGS captures high-frequency appearance details. Recent hybrid methods couple the two representations primarily through depth and normal consistency losses, which is insufficient to resolve the geometric ambiguities inherent to volume rendering. We propose GSNSR, a neural implicit surface framework that leverages 3DGS in two complementary ways to guide SDF optimization, refining global object contours and recovering fine geometric and appearance details. For contour refinement, we introduce a confidence-aware free-space loss that weights pseudo-depth by per-pixel reliability to suppress spurious density in unoccupied regions. For detail recovery, we forgo 3DGS depth as a feature anchor and instead use the ray-surface intersection predicted by the SDF network itself. At this anchor, a decoupled aggregator separately consolidates geometric and appearance attributes from neighboring Gaussians via distance-based weighting, providing the SDF and color networks with non-interfering guidance. Extensive experiments on the DTU and NeRF Synthetic datasets demonstrate that GSNSR achieves highly competitive reconstruction accuracy while significantly suppressing artifacts in challenging scenes.

CCS Concepts

• ***Computing methodologies*** → ***Volumetric models;***

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1. Introduction

Recovering high-fidelity 3D structures from 2D images is a fundamental challenge in computer vision, with broad applications in AR/VR, robotics, and cultural heritage preservation. Among existing approaches, multi-view 3D surface reconstruction has attracted significant attention for its accessibility, requiring only calibrated images rather than specialized scanning hardware. Recent advances in neural implicit representations, such as signed distance functions (SDFs) optimized via volume rendering, have demonstrated remarkable ability to reconstruct smooth, topologically consistent surfaces. However, these methods still struggle with geometric ambiguities in complex or sparsely observed regions, motivating the need for stronger geometric priors.

A pioneering work, NeRF [MST*20], harnesses differentiable volume rendering to achieve novel view synthesis. To enable more precise geometric recovery, NeuS [WLL*21] integrates the Signed Distance Function (SDF) into this framework, achieving high-fidelity surface reconstruction by ensuring geometric consistency during the rendering process. Nevertheless, NeuS suffers from slow convergence due to inefficient position encoding, often extending training times to several hours. Recently, 3DGS [KKLD23] has emerged as a compelling alternative that represents scenes through a collection of anisotropic Gaussians, enabling significantly faster rendering via GPU-accelerated rasterization. However, 3DGS is inherently designed for novel view synthesis rather than surface reconstruction, and its discrete nature makes it ill-suited for precise surface reconstruction.

Implicit SDFs excel at surface topology while 3DGS captures high-frequency appearance details. This complementary relationship motivates a hybrid paradigm that couples 3DGS with implicit SDF representations. Representative works [LSH*24, YLX*24] demonstrate that jointly optimizing these two representations can accelerate convergence and improve reconstruction fidelity compared to SDF-only baselines. These methods commonly adopt rendered depth and normal maps from 3DGS as geometric constraints, supervising the SDF network to align with the Gaussian-derived geometry. However, such coupling operates primarily at the loss-function level, leaving the fundamental ambiguity of volume rendering unresolved. As a result, these methods remain vulnerable to the quality of 3DGS geometry. If the Gaussians fail to converge to accurate surfaces, the erroneous depth and normal cues propagate into SDF optimization, potentially misleading rather than benefiting reconstruction.

Building upon this motivation, we propose GSNSR, a hybrid neural surface reconstruction framework that leverages 3DGS as a geometric and photometric prior to drive SDF optimization. Our central observation is that volume rendering provides no explicit supervision over unoccupied space, so the network readily converges to spurious density and floaters, especially in sparsely observed regions. We introduce a confidence-aware free-space loss that turns the 3DGS-rendered pseudo depth into an explicit free-space prior, enforcing that rays accumulate no opacity before reaching the surface localized by the pseudo depth. This directly attacks the volume-rendering ambiguity that loss-level depth and normal consistency leave unresolved. Crucially, because 3DGS depth is not unconditionally reliable, we design a per-pixel confidence score that combines local depth support, local stability, and view-dependent geometric reliability, and convert it into per-pixel loss weights. This imposes stricter free-space constraints where the pseudo depth is trustworthy while relaxing them in uncertain regions, so the constraint suppresses artifacts while largely preventing unreliable depth from injecting erroneous geometry.

To recover fine geometric and appearance detail, we further fuse 3DGS features into the SDF network at the architectural level. The key difficulty here is where and how the discrete Gaussian features are injected into the continuous field. Prior fusion methods anchor the injection at the 3DGS-rendered depth, so any error in that depth directly displaces the anchor and corrupts the local geometry. Instead, we localize the anchor from the SDF field itself via the ray-surface intersection implied by a sign change along the ray; this severs the dependence on unreliable Gaussian depth and lets the anchor sharpen together with the geometry network during training. At this anchor, rather than hard-replacing the nearest sample's feature, we adopt a distance-based fusion strategy that aggregates neighboring Gaussians and redistributes the result to the two samples bracketing the intersection by their distance to it, supplying spatially grounded guidance. To deliver geometry and appearance cues without interference, the aggregated features are routed through separate geometry and color pathways.

In summary, our main contributions are as follows:

- We introduce a confidence-aware free-space loss that converts 3DGS-rendered pseudo depth into an explicit, per-pixel reliability-weighted free-space prior, directly suppressing floaters and spurious density.
- We propose an SDF-guided fusion scheme that derives the feature-injection anchor from the ray-surface intersection, and redistributes the aggregated Gaussian features to the samples by distance, yielding faithful guidance while preserving representational continuity. Geometry and appearance attributes are routed through separate pathways to avoid cross-interference.
- Evaluations on the DTU and NeRF Synthetic datasets demonstrate that GSNSR achieves competitive surface reconstruction accuracy while effectively reducing artifacts in challenging scenes.

2. Related Work

2.1. 3D neural surface reconstruction

NeRF [MST*20] pioneered differentiable volume rendering for neural implicit representations, enabling high-quality novel view synthesis from multi-view images. Subsequent works have extended this paradigm in complementary directions. Mip-NeRF [BMT*21] mitigates multi-scale aliasing artifacts through integrated positional encoding over conical frustums, while Instant-NGP [MESK22] introduces multi-resolution hash encoding to dramatically accelerate training convergence. Despite these advances in rendering fidelity and efficiency, NeRF-based methods model scene geometry through a volumetric density field that lacks explicit surface correspondence; meshes extracted via density thresholding are therefore typically noisy and topologically ill-defined, motivating subsequent research into surface-aware neural rendering.

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Existing surface reconstruction approaches can be mainly categorized into volume rendering methods based on Neural Implicit Fields and explicit representation methods based on 3DGS [WYZ*24, CW24]. Combining volume rendering with Signed Distance Fields (SDF) has proven effective for acquiring high-fidelity geometric models in implicit reconstruction. NeuS [WLL*21] formulates surface rendering via an unbiased SDF-based volume density, while VolSDF [YGKL21] models volume density as a transformed SDF, establishing a principled connection between the two representations and enabling accurate surface localization through tighter control over the density-to-geometry mapping. However, NeuS often faces slow convergence challenges stemming from positional encoding. Several follow-up works have sought to address this limitation from different angles. NeuS2 [WHH*23] replaces positional encoding with multi-resolution hash grids, achieving significantly faster convergence while maintaining reconstruction quality comparable to the original NeuS. Instant-NSR [ZJY*22] integrates hash grids with SDF, specifically tailoring the framework for human performance reconstruction. Although utilizing occupancy grids facilitates efficient sampling and pruning, optimization is prone to local minima, particularly in regions with occlusion or uneven lighting. Beyond convergence speed, reconstruction accuracy also demands further improvement. Neuralangelo [LME*23] leverages multi-resolution hash encoding alongside numerical gradients for higher-order smoothness regularization, enabling the recovery of large-scale scenes with fine-grained surface details.

3DGS [KKLD23] has gained attention for its impressive novel-view synthesis quality and real-time rendering speed. As a discrete explicit representation, it encodes scene geometry through a set of anisotropic Gaussians, enabling GPU-accelerated rasterization that is orders of magnitude faster than NeRF-based methods. This efficiency makes 3DGS an attractive foundation for surface reconstruction, as it can produce a reasonable geometric scaffold within minutes rather than hours. To extract explicit surfaces from Gaussian primitives, several works have proposed geometry regularization strategies. SuGaR [GL24] introduces a regularization term that encourages Gaussians to align with object surfaces, enabling mesh extraction via Poisson reconstruction. 2DGS [HYC*24] replaces volumetric 3D Gaussians with flat 2D disks, which are inherently better suited for surface fitting and allow more accurate normal estimation. GOF [YSG24] models scene geometry as the level set of a Gaussian opacity field, enabling direct isosurface extraction without post-processing. Despite these advances, 3DGS-based surface reconstruction faces inherent limitations. The discrete nature of Gaussian primitives makes it difficult to enforce global topological consistency, and directly extracted geometries are often prone to noise, holes, and fragmented surfaces in textureless or sparsely observed regions. Furthermore, since 3DGS optimization is primarily driven by photometric loss, geometric accuracy is not guaranteed. Gaussians may adopt arbitrary shapes and orientations that reproduce appearance faithfully while deviating significantly from the true surface. These limitations motivate the integration of 3DGS with implicit representations that offer stronger geometric inductive biases.

2.2. Hybrid representation neural surface reconstruction

Recent hybrid approaches typically employ joint optimization that combines the complementary strengths of 3D Gaussian Splatting (3DGS) and Signed Distance Function (SDF)-based representations. Specifically, 3DGS provides efficient, view-dependent appearance modeling through explicit, rasterization-friendly Gaussian primitives, enabling real-time rendering with high visual fidelity. Meanwhile, the SDF-based neural representation serves as a geometrically rigorous implicit surface prior, enforcing smooth and topologically consistent surface reconstruction [ZLH24, CLWL25, XLC*25]. GSDF [YLX*24] designs a dual-branch system with synchronized optimization, where depth maps rendered by the GS branch narrow and guide the ray-sampling range of the SDF branch, while SDF-predicted geometry provides spatial guidance to regulate the density of 3D Gaussian primitives. This joint supervision enables mutual reinforcement between the two tasks. However, this dual-branch coupling also introduces a fundamental vulnerability. Since 3DGS is inherently an appearance-first rendering framework, its optimizer tends to generate floaters at view-dependent boundaries and high-frequency texture regions to minimize photometric loss. These floaters bear no correspondence to the true object surface yet effectively compensate for rendering residuals. Consequently, the depth maps produced via alpha-blending represent a weighted expectation over all Gaussians along each ray, rather than a faithful measurement of surface depth. When such geometrically unreliable depth estimates are fed back as supervisory signals into the SDF branch, the resulting gradients can systematically bias the zero-level set, giving rise to spurious surfaces or localized geometric collapse. An inherent conflict arises between the photometric constraints of the 3DGS branch and the geometric regularization of the SDF branch. This creates a trade-off where favoring 3DGS supervision yields noisy surfaces, while strict SDF constraints lead to over-smoothed details. One line of research addresses this by imposing stronger geometric regularization on Gaussian primitives to improve their surface alignment, while another explores more principled fusion strategies that go beyond depth supervision, seeking to extract and transfer richer geometric and photometric information latent in the Gaussian representation to the implicit network.

In fact, a well-pretrained 3DGS model inherently encodes rich spatial occupancy and appearance distributions, making it a robust prior for implicit surface reconstruction. SplatSDF [LSD*24] extends Neuralangelo by aggregating the geometric and appearance attributes of Gaussian primitives into high-dimensional feature representations. Rather than applying post consistency constraints, SplatSDF pursues an architecture-level integration of 3DGS and SDF-based NeRF. Specifically, a 3DGS Aggregator extracts and consolidates multiple Gaussian attributes into a compact feature embedding. The depth map rendered by the Gaussian Splatting branch is then used to localize an anchor point along each ray, at which the aggregated 3DGS feature is fused with the SDF feature. This design allows the geometric information encoded in the Gaussian representation to directly inform and guide the SDF network during surface querying. However, SplatSDF still exhibits several limitations that leave room for improvement. Although 3DGS naturally supports a decoupled representation of geometric and appearance attributes, SplatSDF employs a monolithic aggregator

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that jointly encodes both into a unified feature embedding, failing to fully exploit appearance information to guide the optimization of the color network in the SDF-NeRF architecture. Furthermore, while using the 3DGS-rendered depth as the anchor point enables efficient surface localization, it places notable demands on the quality of depth estimation, as noise or artifacts in the rendered depth map can introduce geometric errors into the reconstruction. Since the anchor is fixed at the rendered depth position, the fusion process may become sensitive to inaccuracies in the 3DGS depth estimate.

3. Preliminaries

3.1. 3D implicit neural representation

Given a set of multi-view RGB images with known camera poses, our goal is to reconstruct the high-fidelity 3D scene. The implicit surface $\mathcal{S}$ is defined as the zero-level set of the SDF, given by $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^3 \mid f(\mathbf{x}) = 0\}$.

To render the scene, we cast a ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{v}$ from the camera center $\mathbf{o}$ along the view direction $\mathbf{v}$. We sample N points along the ray, and for each sample point, the color c_i is predicted by the color network. The pixel color $\hat{C}$ is approximated via volume rendering [MST*20]:

$$\hat{C} = \sum_{i=1}^{N} T_i \alpha_i c_i, \quad T_i = \prod_{j=1}^{i-1} (1 - \alpha_j), \tag{1}$$

where T_i denotes the accumulated transmittance and α_i is the discrete opacity of the i-th sample point. Following NeuS [WLL*21], to determine the color contribution of each sample $\mathbf{x}_i$, the discrete opacity α_i is computed from SDF values:

$$\alpha_i = \max\left(\frac{\Phi_s(f(\mathbf{x}_i)) - \Phi_s(f(\mathbf{x}_{i+1}))}{\Phi_s(f(\mathbf{x}_i))}, 0 \right), \tag{2}$$

where $f(\mathbf{x})$ is the signed distance value at point $\mathbf{x}$. $\Phi_s(\cdot)$ is the Sigmoid function, which is parameterized by a learnable parameter s.

3.2. 3D Gaussian Representation

3D Gaussian Splatting (3DGS) [KKLD23] represents a scene as a collection of anisotropic 3D Gaussians $\{\mathcal{G}_k\}_{k=1}^{K}$, where each primitive is parameterized by a set of learnable attributes: a center position $\mu_k \in \mathbb{R}^3$, a full 3D covariance matrix $\Sigma_k \in \mathbb{R}^{3\times3}$ encoding the orientation and extent of the Gaussian, an opacity scalar $\alpha_k \in [0,1]$, and view-dependent appearance represented by spherical harmonic (SH) coefficients. To ensure that Σ_k remains positive semi-definite throughout optimization, it is decomposed as:

$$\Sigma_k = \mathbf{R}_k \mathbf{S}_k \mathbf{S}_k^\top \mathbf{R}_k^\top, \tag{3}$$

where $\mathbf{R}_k$ is a rotation matrix derived from a learnable quaternion, and $\mathbf{S}_k = \mathrm{diag}(s_x, s_y, s_z)$ is a diagonal scaling matrix. Novel views are synthesized by projecting the 3D Gaussians onto the image plane via a differentiable tile-based rasterizer, accumulating color contributions in a front-to-back order weighted by the projected opacity of each Gaussian.

4. Method

We propose GSNSR, which treats a pretrained 3DGS model as a geometric and photometric prior and uses it in two complementary ways to guide SDF optimization within the Instant-NSR-PL [Guo22] framework. The first targets the overall object contour. Because the volume-rendering objective places no constraint on unoccupied space, the SDF network tends to accumulate spurious density in sparsely observed regions and thereby distort the object contour. To counter this, we introduce a confidence-aware free-space loss (Sec. 4.1 and 4.2) that suppresses such density using the 3DGS-rendered pseudo-depth as a soft prior, weighted per pixel by a reliability score so that the penalty is enforced only where the depth can be trusted. At this stage the depth need only indicate, in coarse terms, that the space in front of the surface is empty, so its inaccuracy is tolerable.

The second use targets fine geometric and appearance detail, and fuses 3DGS features into the SDF network at the architectural level, following the principle of SplatSDF [LSD*24] rather than coupling the two representations through loss terms alone. GSNSR is not a simple decoupled variant of SplatSDF, and we return to each of the differences below in detail in Sec. 4.3, 4.4 and 4.5. For aggregator design, where SplatSDF encodes all Gaussian attributes jointly in a single module, GSNSR adopts a Decoupled Aggregator that processes geometric and appearance attributes through separate modules. This design follows the way 3DGS itself separates these two groups of attributes. (Sec.4.3) For anchor localization, SplatSDF places the anchor at the surface position rendered from 3DGS depth, so any error in that depth directly displaces the anchor and injects features at the wrong locations. Anchoring demands a precise position that the noisy 3DGS depth cannot supply. GSNSR therefore derives the anchor from the ray-surface intersection implied by a sign change of the SDF along the ray, which severs the dependence on Gaussian-rendered depth and lets the anchor grow more accurate as the geometry network sharpens during training. (Sec.4.4) For the fusion stage, SplatSDF overwrites the hash encoding of the sample nearest the anchor with the aggregated Gaussian features. GSNSR instead supplies these features through a dedicated auxiliary channel, assigning zero-valued features to samples far from the surface and weighting the samples that bracket the intersection by their distance to the anchor, so the sample closer to the intersection receives a more faithful description of the local geometry than a single hard replacement at the nearest point would provide. (Sec.4.5) The full framework is illustrated in Fig. 2.

4.1. 3DGS Depth Confidence Evaluation

Although 3DGS depth estimates are insufficiently precise to serve as a hard anchor for feature localization, they remain informative as a soft spatial prior. We therefore leverage pseudo-depth for free-space supervision, while mitigating its unreliability through a per-pixel confidence weighting scheme. We posit that a high-quality 3DGS depth observation should satisfy three complementary properties: spatial continuity, local consistency, and view-consistent geometric reliability.

Local Neighborhood Support Score. To assess whether a pixel is supported by surrounding valid depth estimates, we compute the

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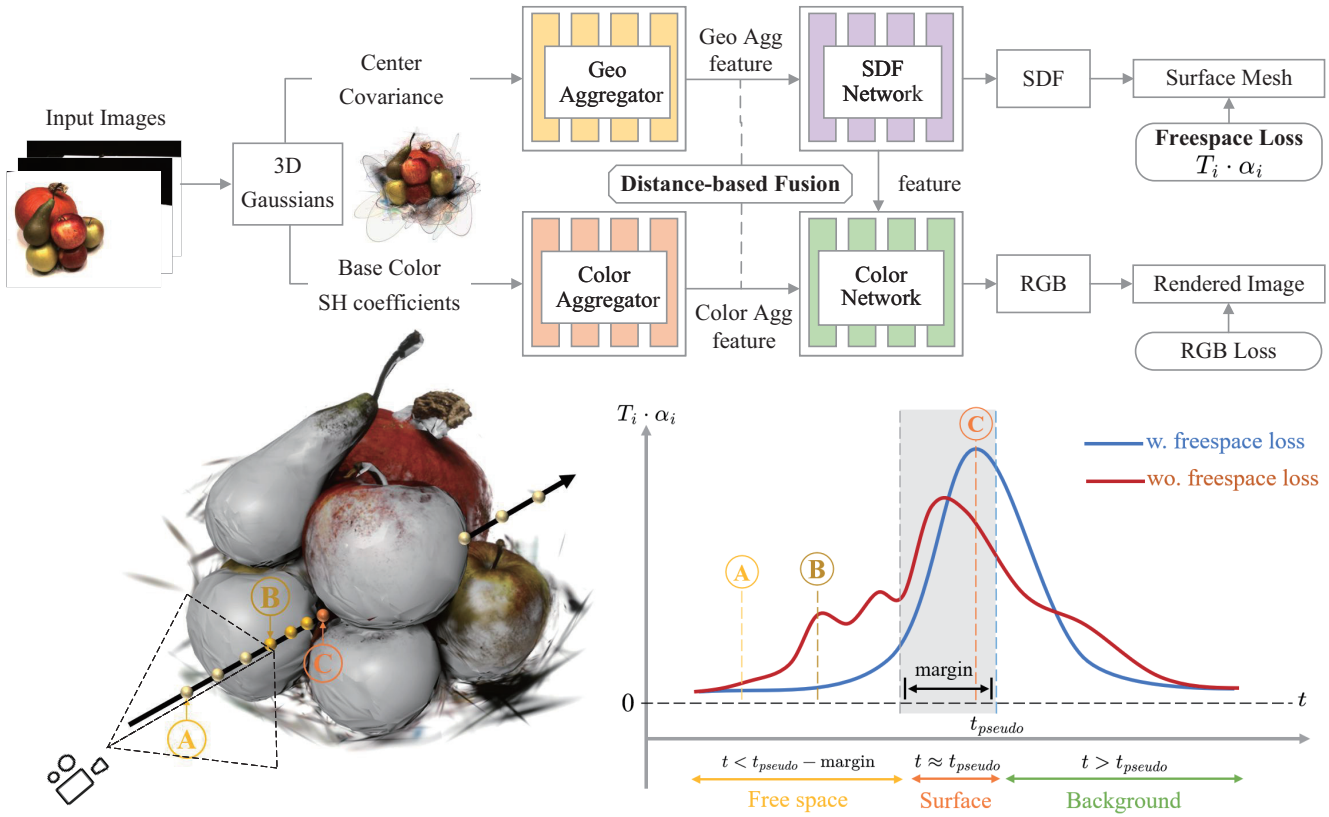


Figure 2: *Overview of the GSNSR architecture. Our framework integrates 3D Gaussian primitives as a prior with a neural implicit field. Decoupled aggregators extract geometric and photometric features from neighboring Gaussians, which are then used to guide the SDF network and the color network, respectively. The final image is synthesized via differentiable volume rendering, enabling high-fidelity surface reconstruction under robust geometric constraints. The lower panel visualizes the effect of the free-space loss: for sampling points A and B, which lie far from the surface, the corresponding weights should remain low to prevent artifacts in empty regions; for the near-surface sample C, its weight should be close to the peak value, as it contributes more to the reconstruction.*

ratio of valid pixels $r(p)$ within a $k \times k$ local window centered at p.

$$S_{\text{supp}}(p) = \min\left(\frac{r(p)}{r_{\text{thresh}}}, 1\right), \tag{4}$$

where r_{thresh} is the minimum ratio threshold. The confidence is evaluated only for pixels within the foreground mask; all other pixels receive $S_{\text{supp}}(p) = 0$.

Local Depth Consistency Score. To capture local depth consistency, we compute the relative standard deviation of depth values within $\mathcal{N}(p)$, normalized by the local mean depth and clipped by a threshold τ_σ to yield $\tilde{\sigma}_{\text{rel}}(p) = \min\left(\frac{\sigma_{\text{rel}}(p)}{\tau_\sigma}, 1\right)$. A large $\tilde{\sigma}_{\text{rel}}$ indicates depth discontinuities or noisy estimates. The consistency score is then defined as:

$$S_{\text{cons}}(p) = [1 - \gamma \cdot \tilde{\sigma}_{\text{rel}}(p)]_0^1, \tag{5}$$

where $[\cdot]_0^1$ denotes clamping to the range $[0, 1]$, γ is a global consistency weight.

View-Dependent Geometric Reliability. Rendered depth maps can appear locally smooth yet be geometrically unreliable at grazing-angle pixels, where the surface normal is nearly perpendicular to the viewing ray and depth varies rapidly in image space. We therefore quantify it via the image-space depth gradient magnitude, computed by Sobel filtering and normalized by the scene median depth for scale invariance. The result is further normalized by the 95% quantile over valid pixels to yield $\tilde{g}(p) \in [0, 1]$. A multiplicative penalty $\exp(-\lambda_g \cdot \tilde{g}(p))$ is then applied to suppress confidence at high-gradient regions, with λ_g controlling the penalty strength.

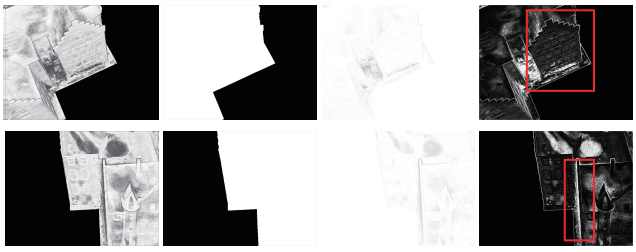
Building on the three properties above, we define the overall confidence score $C(p)$ for each pixel p as their product:

$$C(p) = S_{\text{supp}}(p) \cdot S_{\text{cons}}(p) \cdot \exp\big(-\lambda_g \cdot \tilde{g}(p)\big). \tag{6}$$

S_{supp} assesses local depth support density, S_{cons} penalizes local depth inconsistency, and the exponential term suppresses unreliable estimates at grazing-angle observations.

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As illustrated in Fig. 3 on the DTU scan24 scene, each component of the confidence score captures a distinct and complementary aspect of depth reliability. The support score S_{supp} in (b) reflects the density of neighboring Gaussian contributions. The consistency score S_{cons} in (c) remains broadly high across the scene, indicating that the Gaussian depth estimates are locally stable in this scene. The view-dependent geometric reliability term in (d) most clearly demonstrates its intended behavior: in the frontal viewpoint (top row), the building facade constitutes a large, nearly planar surface oriented toward the camera, yielding low depth gradient norms and correspondingly high confidence across the highlighted region. In contrast, in the oblique viewpoint (bottom row), the same facade becomes highly foreshortened and presents steep depth gradients, causing the exponential term to assign low confidence to these grazing-angle observations. This view-dependent suppression is critical for preventing unreliable depth estimates at oblique surfaces from introducing erroneous geometric supervision into the free-space loss.



(a) Confidence (b) Support Score (c) Consistency Score (d) Depth Grad Norm

Figure 3: *Confidence evaluation on DTU scan24. (a) final confidence; (b) support score; (c) consistency score; (d) depth gradient norm. Brighter is more reliable in (a)–(c), whereas brighter means larger gradient (less reliable) in (d).*

4.2. Confidence-aware free-space constraints

The raw confidence score $C(p)$ is linearly rescaled to $[0,1]$ by clipping values below c_{low} to zero, where c_{low} is the threshold that discards highly unreliable depth estimates. Let $\hat{C}(p)$ denote this rescaled confidence score. For each ray $\mathbf{r}(t) = \mathbf{o} + t\mathbf{v}$, the pseudo depth $\hat{d}$ rendered from 3DGS provides a surface position estimate along the ray. The free-space region $\mathcal{F}$ is defined as the interval from the camera origin up to a position slightly in front of this estimated surface:

$$\mathcal{F} = \left\{ t \;\middle|\; t < \hat{d} - \epsilon \right\}, \tag{7}$$

where ϵ is a small safety margin, accounting for inaccuracies in the 3DGS-rendered depth.

Theoretically, the volume rendering weight accumulated within $\mathcal{F}$ should be zero, since any opacity in this region corresponds to floaters or erroneous geometry ahead of the true surface. For each ray i, we define the free-space mass as the sum of rendering weights over all samples falling within $\mathcal{F}$:

$$m_i = \sum_{j:\, t_{ij} \in \mathcal{F}} w_{ij}, \tag{8}$$

where $w_{ij} = T_{ij}\alpha_{ij}$ is the volume rendering weight of the j-th sample on the ray, computed from the SDF-induced opacity as in NeuS. The confidence-weighted free-space loss is then defined as:

$$\mathcal{L}_{fs} = \frac{\sum_i \hat{C}(p_i) \cdot m_i}{\sum_i \hat{C}(p_i)}, \tag{9}$$

This formulation weights the geometric constraint by depth confidence, focusing the penalty on reliable regions while relaxing it where 3DGS depth estimates are uncertain.

4.3. Decoupled 3DGS Aggregator

3DGS maintains an explicit separation between geometric parameters (mean μ, covariance Σ) and appearance attributes (SH coefficients, opacity α). We mirror this structure in the implicit field by introducing two dedicated aggregators that process geometry and appearance along separate pathways.

For a selected Gaussian primitive G_i, we apply hash encoding to its center μ_i using the same encoding strategy as for sample points. The encoded position together with the covariance Σ_i is fed into the geometry aggregator to yield $f_{\text{geo}}^{(i)}$. For appearance, we decompose the SH coefficients into a view-independent base color c_i and view-dependent coefficients SH_i, which the color aggregator maps to $f_{\text{col}}^{(i)}$:

$$\mathbf{f}_{\text{geo}}^{(i)} = \mathcal{A}_{geo}\left(h(\mu_i), \Sigma_i\right), \tag{10}$$

$$\mathbf{f}_{\text{col}}^{(i)} = \mathcal{A}_{col}\left(\mathbf{c}_i, SH_i\right), \tag{11}$$

where $h(\cdot)$ is the hash encoding and $\mathcal{A}_{geo}$, $\mathcal{A}_{col}$ the two aggregators. Routing the attribute groups to separate networks lets the color network receive dedicated view-dependent photometric guidance for faithful appearance modeling, while keeping appearance signals from contaminating the geometric optimization of the SDF network.

4.4. SDF-Guided Anchor Localization

The anchor determines where 3DGS features are injected into the implicit field, so it must sit precisely on the surface; an anchor displaced from the true surface injects Gaussian features at the wrong location and corrupts the local geometry. The 3DGS-rendered depth is too unreliable for this purpose, near fine structures and at grazing-angle observations its error is large, so we do not use it to place the anchor. Instead, we localize the anchor from the SDF field itself.

Consider two adjacent sampling points p_j and p_{j+1} along a ray. A surface crossing is signaled by a sign change in their SDF values, $f(p_j) \cdot f(p_{j+1}) < 0$. Within such an interval, we estimate the intersection x^* by linear interpolation of the SDF values:

$$\mathbf{x}^* = \frac{f(\mathbf{p}_j)\mathbf{p}_{j+1} - f(\mathbf{p}_{j+1})\mathbf{p}_j}{f(\mathbf{p}_j) - f(\mathbf{p}_{j+1})}. \tag{12}$$

Localizing the anchor this way keeps it consistent with the SDF being optimized, free of any external depth signal that could conflict with the network's own surface prediction. It also lets the anchor improve together with the geometry. As the SDF network sharpens during training, the interpolated intersections move closer to

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the true surface, so a better geometry yields a more precise anchor, which in turn delivers more faithful 3DGS guidance.

4.5. Hybrid Feature Fusion

We implement the distance-based feature fusion strategy, as illustrated in Fig. 4. We perform a radius-bounded K-Nearest Neighbor (KNN) search [XXP*22] to retrieve neighboring Gaussian primitives.

$$\omega(\mathbf{x}^*, G_i) = \exp\left(-\frac{1}{2}(\mathbf{x}^* - \mu_i)^\top \Sigma_i^{-1}(\mathbf{x}^* - \mu_i)\right), \quad (13)$$

$$\mathbf{F}_{geo}(\mathbf{x}^*) = \frac{1}{K}\sum_{G_i \in \mathcal{K}(\mathbf{x}^*)} \alpha_i \cdot \omega(\mathbf{x}^*, G_i) \cdot \mathbf{f}_{geo}^{(i)}, \quad (14)$$

$$\mathbf{F}_{col}(\mathbf{x}^*) = \frac{1}{K}\sum_{G_i \in \mathcal{K}(\mathbf{x}^*)} \alpha_i \cdot \omega(\mathbf{x}^*, G_i) \cdot \mathbf{f}_{col}^{(i)}, \quad (15)$$

where K is the number of contributing neighbors in the search radius, and α_i denotes the opacity of G_i. The spatial weight $\omega(\mathbf{x}^*, G_i)$ is determined by the anisotropic Gaussian kernel using the Mahalanobis distance.

For conciseness, we uniformly denote the aggregated geometry feature $\mathbf{F}_{geo}(\mathbf{x}^*)$ and color feature $\mathbf{F}_{col}(\mathbf{x}^*)$ as $\mathbf{F}_{gs}(\mathbf{x}^*)$. To align this intersection-based feature with the ray sampling points, we distribute $\mathbf{F}_{gs}(\mathbf{x}^*)$ back to $\mathbf{p}_j$ and $\mathbf{p}_{j+1}$ using distance-based weighting. The final injected features are defined as:

$$\mathbf{v}_j = \frac{d_2}{d_1 + d_2}\mathbf{F}_{gs}(\mathbf{x}^*), \quad \mathbf{v}_{j+1} = \frac{d_1}{d_1 + d_2}\mathbf{F}_{gs}(\mathbf{x}^*), \quad (16)$$

where $d_1 = |t_j - t^*|$ and $d_2 = |t_{j+1} - t^*|$ represent the distances along the ray from the sampling depths t_j and t_{j+1} to the estimated surface depth t^*. This design ensures that the closer a sampling point is to the potential surface, the stronger the Gaussian prior guidance it receives. The injected features $\mathbf{v}_j$ and $\mathbf{v}_{j+1}$ maintain the same shape as the full set of ray sampling points and are provided as auxiliary inputs to both the geometry and color networks alongside the standard hash-encoded features. For all remaining sampling points that either lack nearby Gaussian primitives within the search radius or are not adjacent to any sign-change interval, the corresponding injected features are set to zero. This design supplies spatially grounded 3DGS guidance near potential surfaces while preserving the continuity of the coordinate-based representation.

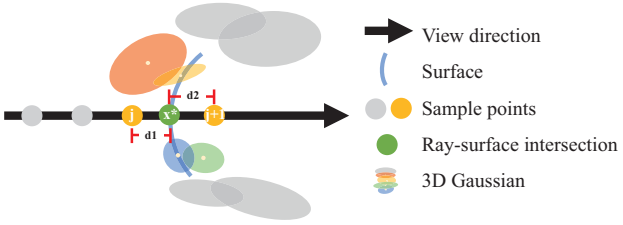


Figure 4: *The Distance-based fusion strategy. At the ray-surface intersection $\mathbf{x}^*$, we aggregate features from K nearest Gaussians based on spatial proximity. These features are then propagated to adjacent sample points p_j and p_{j+1} based on d_1 and d_2.*

4.6. Losses

The overall training objective is defined as a weighted combination of photometric reconstruction and geometric regularizations:

$$\mathcal{L} = \mathcal{L}_{rgb} + \lambda_1\mathcal{L}_{eik} + \lambda_2\mathcal{L}_{mask} + \lambda_3\mathcal{L}_{curv} + \lambda_4\mathcal{L}_{fs}, \quad (17)$$

where $\mathcal{L}_{rgb}$ denotes the L_1 photometric loss between rendered and ground-truth pixels, and $\mathcal{L}_{eik}$ enforces the Eikonal constraint on the SDF gradients. $\mathcal{L}_{mask}$ supervises the accumulated ray opacity $\hat{O}(r)$ against the binary foreground mask via binary cross-entropy. $\mathcal{L}_{curv}$ penalizes the mean absolute Laplacian of the SDF at points sampled within the scene bounding sphere, encouraging smooth surface reconstructions, as in [LSD*24]. $\mathcal{L}_{fs}$ is the free-space loss. The coefficients $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are hyperparameters balancing the contribution of each term.

5. Experiment

5.1. Experimental settings

Datasets. Following prior work [LSD*24], we conduct experiments on 12 selected scenes from the DTU dataset [JDV*14]. Each scene comprises 49 to 64 images captured by a robot-mounted monocular RGB camera. The dataset features a diverse range of materials and complex geometries. Each scene is evaluated using the foreground masks as provided in 2DGS [HYC*24]. We additionally evaluate on the NeRF Synthetic dataset [MST*21]. Each scene provides 100 training images and 200 test images at a resolution of 800×800 pixels, rendered from upper-hemisphere viewpoints with a white background.

Implementation Details. We adopt a coarse-to-fine multi-resolution hash encoding [MESK22, LME*23] with 16 levels, whose grid resolutions grow geometrically from 2^5 to 2^{11} and whose per-level hash table is capped at 2^{22} entries. Each level uses 2 features per hash entry, which we found sufficient for our object-centric scenes while keeping training cost low. For the proposed Decoupled Aggregators, the output dimensions of geometric and color features are set to 32 and 64, respectively. Following SplatSDF [LSD*24], we set the number of neighboring Gaussians $K = 4$ for feature aggregation. A coarse-to-fine optimization strategy is adopted to improve convergence stability. To ensure stable ray-surface intersection during volume rendering, the aggregators are activated after 10k iterations, by which point the SDF network has established a reliable initial geometry and intersection points can be reliably located. For the pseudo-depth confidence map, the neighborhood support threshold is set to $r_{\text{thresh}} = 0.85$ in Eq.4. The consistency weight and relative standard deviation threshold are set to $\gamma = 0.25$ and $\tau_\sigma = 0.03$ in Eq.5. The depth gradient penalty is set to $\lambda_g = 1.0$ in Eq.6. The loss weights follow those of the baseline, with $\lambda_1 = 0.1$, $\lambda_2 = 0.1$, $\lambda_3 = 0.0005$ and $\lambda_4 = 0.3$. The safety margin ϵ is adaptively configured based on the depth scale of the scene: it is set to 20% of the bounding sphere radius for the DTU dataset to account for geometric noise, and 10% for the NeRF Synthetic dataset owing to its more accurate pseudo-depth priors. All experiments are conducted on a single NVIDIA RTX 4090 GPU. The model is trained for 300k iterations, requiring approximately 8 hours per scene.

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Table 1: *Quantitative Comparisons on the DTU Dataset. (Chamfer distance, CD ↓). Pink and yellow denote the best and second-best results. Our method achieves the best overall performance among all baseline geometric reconstruction methods.*

	Scan	24	37	40	55	63	65	69	83	105	106	110	114	Mean
SDF-based Methods	Instant-NSR-PL	2.86	2.81	2.09	0.81	1.65	1.39	1.47	1.67	1.12	1.22	2.3	0.98	1.7
	NeuS	1	1.37	0.93	0.43	1.1	0.65	0.57	1.48	0.83	0.52	1.2	0.35	0.87
	NeuS2	0.56	0.76	0.49	0.37	0.92	0.71	0.76	1.22	0.63	0.59	0.89	0.4	0.69
	Geo-NeuS	0.46	0.85	0.38	0.43	0.89	0.5	0.5	1.26	0.66	0.52	0.82	0.31	0.63
	Neuralangelo	0.37	0.72	0.35	0.35	0.87	0.54	0.53	1.29	0.73	0.47	0.74	0.32	0.61
	SplatSDF	0.35	0.67	0.32	0.31	0.84	0.54	0.55	1.16	0.65	0.47	0.75	0.31	0.58
	Ours	0.42	0.67	0.31	0.37	0.71	0.54	0.53	1.05	0.55	0.49	0.59	0.38	0.55
3DGS-based Methods	SuGaR	1.47	1.33	1.13	0.61	2.25	1.71	1.15	1.63	1.07	0.79	2.45	0.98	1.38
	GaussianSurfels	0.66	0.93	0.54	0.41	1.06	1.14	0.85	1.29	0.79	0.82	1.58	0.45	0.88
	2DGS	0.48	0.91	0.39	0.39	1.01	0.83	0.81	1.36	0.76	0.7	1.4	0.4	0.79
	MVS-GS	0.43	0.79	0.33	0.35	0.91	0.67	0.55	1.28	0.61	0.56	0.77	0.35	0.63
Hybrid Methods	GSDF	0.59	0.94	0.46	0.38	1.3	0.77	0.73	1.59	0.76	0.59	1.22	0.38	0.81
	3DGSR	0.68	0.84	0.7	0.39	1.16	0.87	0.77	1.48	0.87	0.69	0.8	0.42	0.81
	GS-Pull	0.51	0.56	0.46	0.39	0.82	0.67	0.85	1.37	0.73	0.54	1.39	0.35	0.72

Table 2: *Quantitative Comparisons on the DTU Dataset.(PSNR ↑) Pink and yellow denote the best and second-best results. Our method demonstrates superior rendering quality, yielding the best results in half of the tested scenes and achieving the second-best overall performance in terms of average PSNR.*

Scan	24	37	40	55	63	65	69	83	105	106	110	114	Mean	Time
Instant-NSR-PL	23.86	24.97	25.3	25.43	29.52	26.17	22.93	26.72	27.71	23.12	25.44	26.7	25.66	8.5min (30k)
NeuS2	28.44	27.14	29.7	29.67	31.75	27.83	24.84	31.24	30.57	26.05	28.93	28.98	28.76	5min(15k)
NeuS	26.62	23.64	26.43	25.59	30.61	32.83	29.24	33.71	31.97	32.18	28.92	28.41	29.18	8h
SplatSDF	29.63	30.39	29.75	28.44	31.24	28.74	29	29.61	29.41	28.83	29.45	28.68	29.43	12h
Neuralangelo	30.64	27.78	32.7	34.18	35.15	35.89	31.47	36.82	35.92	36.61	32.6	31.2	33.41	14h
Ours	33.37	32.05	33.8	34.89	35.3	31	29.64	33.39	32.78	27.21	30.12	32.71	32.19	8h

5.2. Comparisons

Quantitative Evaluation. We select Instant-NSR-PL [Guo22], a PyTorch-Lightning implementation that provides object-level DTU results, as our primary baseline for its remarkable efficiency. It is distinct from the original Instant-NSR [ZJY*22], which targets human-performance capture and does not evaluate on DTU. To demonstrate the efficacy of our approach, we conduct a comprehensive comparison against several state-of-the-art methods, including SDF-based methods (Instant-NSR-PL [Guo22], NeuS [WLL*21], NeuS2 [WHH*23], Geo-NeuS [FXOT22], Neuralangelo [LME*23], SplatSDF [LSD*24]) and 3DGS-based methods (SuGaR [GL24], GaussianSurfels [DXX*24], 2DGS [HYC*24], MVS-GS [KPL25]). The quantitative results of Instant-NSR-PL are directly cited from NeuS2. The experimental results in Table 1 demonstrate that our method achieves highly competitive performance across the majority of scenarios, with substantial gains observed in scan63, scan83 and scan110. Our approach effectively captures intricate object details while maintaining overall geometric integrity.

Table 2 reports a quantitative comparison of PSNR and training time across methods on the DTU dataset. Our method attains the highest PSNR on scans 24, 37, 40, 55, 63 and 114 and ranks second overall, with a mean PSNR of 32.19 dB, indicating competitive rendering quality. Although Neuralangelo achieves the best mean PSNR (33.41 dB), it requires roughly 14 hours of training; by contrast, our method reaches comparable quality in only 8 hours, reducing training time by 33% relative to SplatSDF. Overall, these results indicate that our method offers a favorable trade-off between rendering fidelity and training efficiency, outperforming most baselines under comparable or shorter training budgets.

Qualitative Evaluation. Fig. 5 presents qualitative comparisons on the DTU dataset. Across diverse object types, GSNSR consistently produces clean and detail-rich reconstructions. For scan 24, GSNSR recovers sharper window structures but shows slight concavities in the shadowed region. Our confidence mechanism correctly flags this region as unreliable and relaxes the free-space constraint accordingly. However, soft confidence weighting attenuates rather than fully removes the influence of an entirely erroneous 3DGS depth prior, leaving a slight inward bias that methods without a free-space term do not exhibit. We regard this as an acceptable trade-off given the accuracy gains GSNSR achieves elsewhere. For scan 65, the nasal-cavity region suffers from severe photometric ambiguity due to insufficient lighting. GSNSR and Neuralangelo reconstruct a well-defined, artifact-free surface in this region, whereas 2DGS and SplatSDF exhibit pronounced surface noise and missing geometry in the highlighted area. Within the eye socket, GSNSR recovers the original cavity structure, while the other methods all produce artifacts there. For scan 83, the doll's

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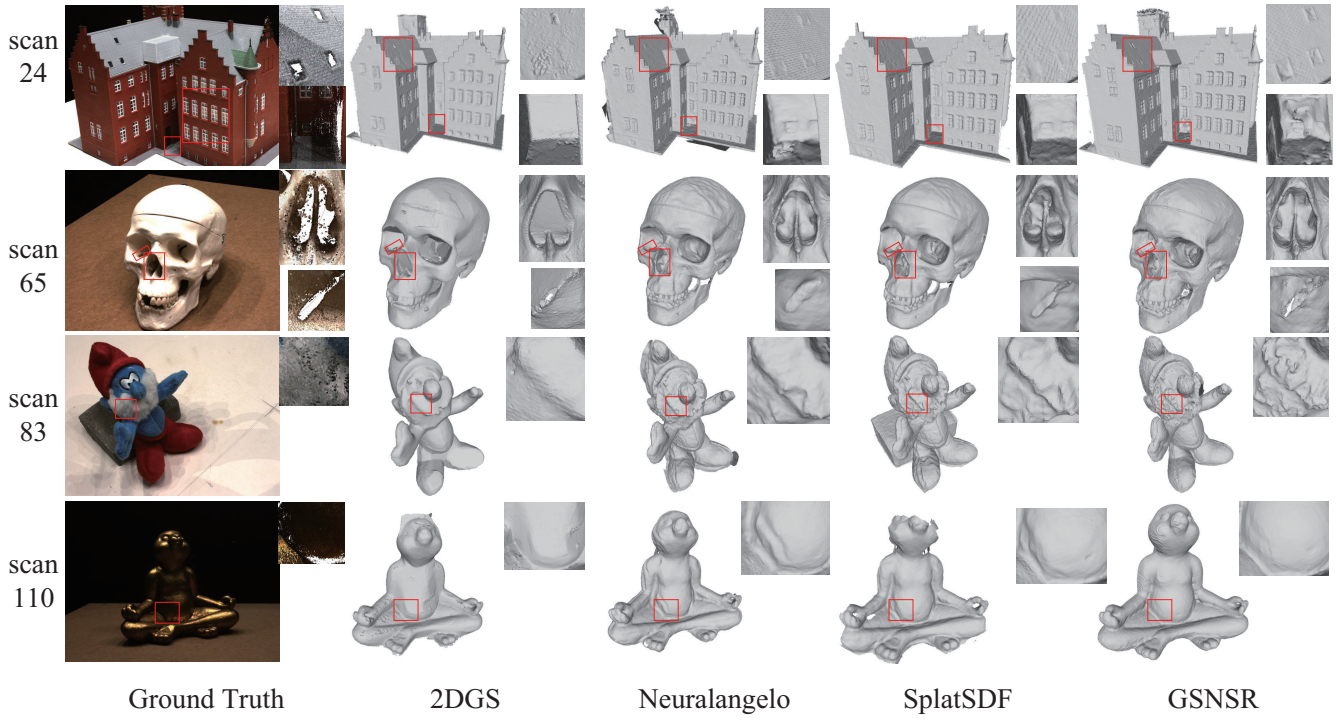


Figure 5: *Qualitative comparison on the DTU dataset.*

beard represents high-frequency geometric detail. GSNSR shows a clear advantage in recovering such fine structures, producing more complete and detailed beard geometry, whereas Neuralangelo and SplatSDF yield overly smoothed results and 2DGS exhibits fragmented surfaces. For scan 110, GSNSR restores sharp object boundaries and smooth surface contours, while 2DGS produces rougher surface detail and both Neuralangelo and SplatSDF show concavities in the torso region.

5.3. Ablation Study

To validate the contribution of each proposed component, we conduct an ablation study with four configurations. Table 3 presents the quantitative ablation results on the scan63, Lego and Mic scenes. For each variant, we report Chamfer distance and PSNR. This allows us to track how each component affects both geometric accuracy and rendering quality.

Our experiments use Instant-NSR-PL [Guo22] as the baseline. The SplatSDF-style variant retains the free-space loss but otherwise follows the fusion design of SplatSDF. It adopts the same aggregator as SplatSDF and takes the 3DGS-rendered depth as the anchor point for surface localization. The 3DGS feature is still injected as an auxiliary input aligned with the ray samples. We place the aggregated feature directly at the position of the surface sample point and set all other positions to zero, instead of distributing it to neighboring samples by distance. The w/o free-space variant shares the same aggregator design as GSNSR and differs only in omitting the free-space loss. The w/o decouple variant uses the same aggregator as the SplatSDF-style variant, while all other modules remain identical to GSNSR. Finally, the full GSNSR model incorporates the free-space loss, employs the decoupled aggregator, uses ray-surface intersection points as anchors for surface localization, and adopts distance-based weighting for fusion. The results reveal the contribution of each proposed component, with the full model achieving the best Chamfer distance across all scenes.

As shown in Fig. 6, for scan63, the scene consists of several smooth, closely arranged fruits. The depression becomes progressively smaller from the SplatSDF-style variant through the w/o free-space and w/o decouple variants, and GSNSR produces the most complete surface, smoothly closing the concavity and recovering a contour closest to the ground truth. The SplatSDF-style variant anchors feature aggregation at the 3DGS-rendered depth and directly replaces the feature, so geometric guidance is injected at an inaccurate location and biases the SDF zero-level set inward, producing the most pronounced depression. By relocating the anchor to the SDF-predicted ray-surface intersection and distributing the aggregated feature to neighboring samples by distance, our fusion strategy preserves surface continuity, substantially reducing the depression. The decoupled aggregator further prevents appearance cues from leaking into the geometry branch. This produces the most complete and smooth surface.

For Lego, which contains intricate mechanical structures with thin, repetitive geometry, the SplatSDF-style variant produces clear geometric errors on the wheels and treads and yields the lowest PSNR. Anchoring at the unreliable 3DGS depth and directly replacing the feature there injects corrupted geometric guidance in these regions, degrading both geometry and rendering. The w/o

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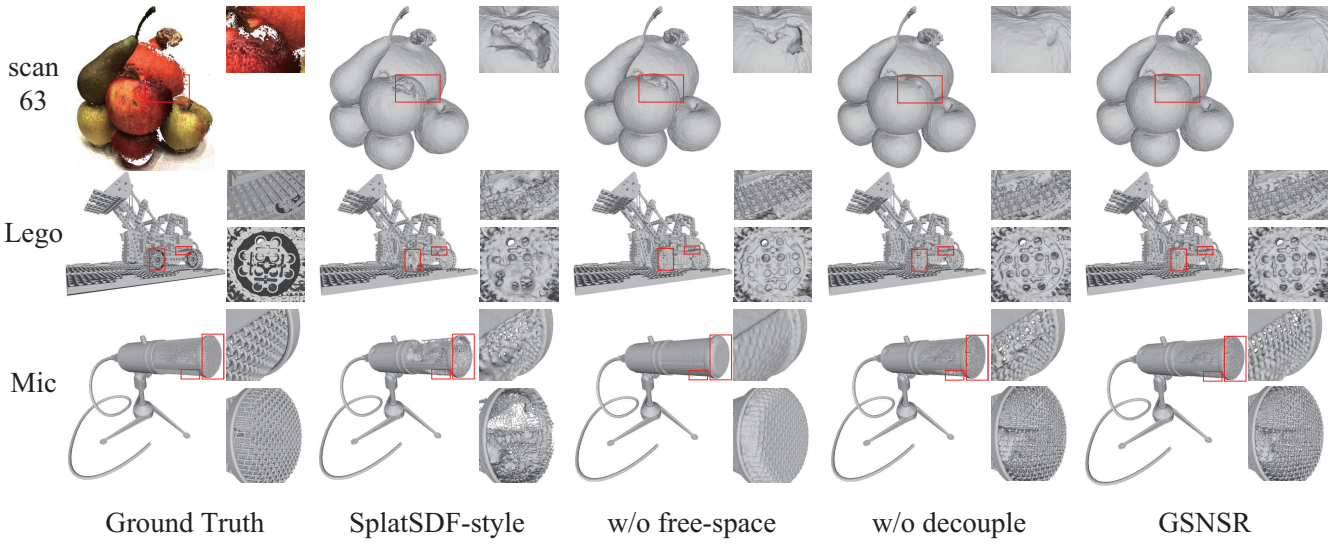


Figure 6: *Ablation study on the impact of the proposed components on scan63, 'Lego' and 'Mic'.*

Table 3: *Ablation results on scan63, 'Lego' and 'Mic'. Pink and yellow denote the best and second-best results. GSNSR achieves the best Chamfer distance across all three scenes.*

Variant	free-space loss	aggregator	anchor points	fusion	scan63		Lego		Mic	
					CD↓	PSNR↑	CD↓	PSNR↑	CD↓	PSNR↑
SplatSDF-style	√	couple	3DGS depth	replace	0.73	35.27	1.1	32.56	0.77	30.25
w/o free-space	×	decouple	intersection	distance-based	0.72	35.35	1.34	35.54	0.56	35.18
w/o decouple	√	couple	intersection	distance-based	0.72	35.19	1.08	35.12	0.5	35.39
GSNSR	√	decouple	intersection	distance-based	0.71	35.3	1.07	35.24	0.48	35.45

free-space variant recovers a more complete overall shape but introduces floater artifacts, most evident beneath the chassis. These floaters help fit the photometric signal and give it the highest PSNR, yet they corrupt the geometry and result in the worst Chamfer distance, showing how the absence of the free-space loss trades geometric accuracy for appearance fitting. With the free-space loss reintroduced, the w/o decouple variant suppresses these floaters, leaving only minor depressions in fine details. Its PSNR nonetheless remains lower than that of GSNSR, isolating the benefit of the decoupled aggregator, which provides the color network with dedicated photometric guidance. GSNSR achieves the best Chamfer distance and the cleanest reconstruction, recovering the mechanical details while keeping the surface free of floaters.

For Mic, which features high-frequency lattice structures with open cavities, the SplatSDF-style variant produces severe geometric errors on the microphone head, causing this structure to collapse almost entirely and yielding the worst Chamfer distance and PSNR. This failure originates from an uncritical reliance on the 3DGS depth. When erroneous anchor placement injects the aggregated features at incorrect spatial locations, they conflict with the hash encoding and destabilize the geometry network's internal representations. Critically, the free-space loss also relies on the 3DGS depth to define the unoccupied region along each ray. Because the SplatSDF-style variant ties both the anchor localization and the free-space supervision to this same unreliable signal, the resulting errors become mutually reinforcing, and this compounding effect drives the observed collapse. The w/o free-space variant recovers a more complete overall shape but fails to reconstruct the open lattice, as spurious density fills the cavities that should remain hollow. With the free-space loss reintroduced, the w/o decouple variant suppresses this cavity-filling and produces a more complete surface, leaving only minor depressions in fine details. As in Lego, its PSNR remains lower than that of GSNSR, isolating the benefit of the decoupled aggregator. GSNSR achieves the best Chamfer distance and the best PSNR, cleanly recovering the fine open lattice while suppressing spurious volume.

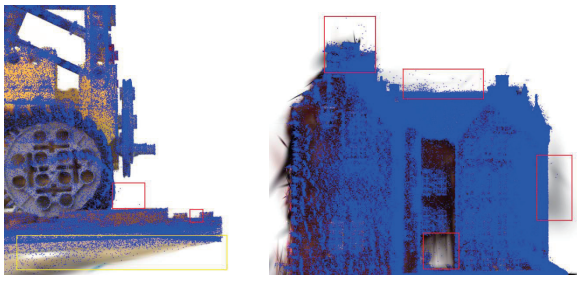
5.4. Limitation and Future Works

Although we introduce a confidence-aware mechanism to evaluate the reliability of Gaussian depth, our framework consumes this depth as a soft prior and does not correct it. The confidence weighting down-weights unreliable depth wherever its failure is detectable from local support, local stability, or viewing geometry, yet it can only attenuate the influence of an erroneous prior. Where the Gaussian depth is severely wrong, the residual free-space constraint still leaves a slight geometric bias that weighting alone cannot fully re-

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move. The method therefore also struggles on scenes with excessive noise, where low-confidence depth priors may insufficiently suppress geometric artifacts. Our current framework builds upon vanilla 3DGS, whose geometric accuracy is inherently limited. As illustrated in Fig. 7, Gaussians reconstructed on DTU scenes exhibit a notable proportion of floaters detached from the actual surface, whereas those on NeRF Synthetic scenes conform more tightly to the object geometry. This discrepancy directly affects the reliability of the pseudo-depth priors used in our free-space loss, and partially explains the relatively weaker artifact suppression observed on DTU. We believe that integrating more geometrically faithful Gaussian representations would substantially benefit our pipeline.

Our method exhibits suboptimal reconstruction quality on large-scale, unbounded scenes such as those in the Tanks and Temples dataset. We attribute this primarily to the limited representational capacity of our current network architecture, which was originally designed and optimized for object-centric, bounded scenes. In such settings, the MLP-based SDF network can adequately cover the scene's geometric complexity within a compact volume. However, unbounded outdoor scenes introduce significantly greater spatial extent, depth range ambiguity, and viewpoint diversity, which collectively exceed the expressive capacity of the current network. The fixed network size struggles to simultaneously capture both fine-grained local surface details and the global geometric layout of large scenes, leading to incomplete or inaccurate reconstructions in distant regions. We hope that future work will explore scene contraction strategies, spatial partitioning, or hierarchical network designs to scale the model's representational capacity to unbounded environments, enabling more accurate SDF learning across the full scene extent.



(a) Lego (b) scan24

Figure 7: *Limitation conditions and results.*

6. Conclusion

In this paper, we present GSNSR, a hybrid neural surface reconstruction framework that harnesses 3DGS as a geometric and photometric prior to guide neural SDF optimization. To address geometric artifacts and refine object contours, we introduce a confidence-aware free-space loss that leverages 3DGS-rendered pseudo depth. By converting this pseudo depth into an explicit free-space prior weighted by per-pixel reliability, this loss effectively suppresses spurious floaters and density in unoccupied regions, directly optimizing the overall object boundaries. To recover fine geometric and appearance details, we propose a novel feature injection strategy that departs from unreliable 3DGS-rendered depth and localizes the feature-injection anchor at the ray-surface intersection predicted by the SDF network itself. Building upon this anchor, a decoupled aggregation mechanism separates geometric and appearance attributes from neighboring Gaussians and propagates them to the intersecting samples via distance-based weighting. Extensive experiments on the DTU and NeRF Synthetic datasets demonstrate that GSNSR achieves competitive reconstruction accuracy with significantly richer geometric details and fewer artifacts across challenging scenes.

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